CrashGame

Betsoft Gaming

Version 1.0

1. Lobby Protocol

This document describes the communication protocol between client and the server in the “Crash game” multiplayer game.
Interaction is carried out via websockets in JSON format.
The startGameUrl and session id of the logged-in user are passed to the client through the http request parameters: startGameUrl (wss://host:port/websocket/crash), sessionId.
Please note, that we recommend to use only secure websocket connections (wss://).
Rooms can be created and deleted by the server as needed, depending on the current number of players.
Type of all identifiers (id, rid, roomid, ..) is long. As some features are game-specific, related fields in other games where not applicable should be ignored.

1.1 Error handling

In response to any client request may come an error message like this:

```
{
  "code": 1,
  "msg": "Internal error",
  "date": 1496748898812,
  "class": "Error",
  "rid": 1
}
```

```
{
  "code": 2,
  "msg": "Server shutdown",
  "date": 1496748898812,
  "class": "Error",
  "rid": -1
}
```

Request parameters:

code

error code

msg

error description (may be localized)

date

datetime in msec. from 01.01.1970 (standart unix timestamp)

class

message class

rid

request Id, type is int. Optional parameter, if not -1 this parameter contains id of original request.

Possible error ranges

1-999

FATAL ERROR

1000-4999

ERROR

5000-9999

WARNING

Full list of error codes could be found in section 4

Handling 'FATAL ERROR':

Display message to user and close lobby.

Handling Lobby Websocket errors:

In case of websocket error, you must re-establishing websocket connection to lobbyUrl and send 'Enter Lobby' request.

1.2 Enter Lobby

This request is a first request which should be send by the client after establishing websocket connection to server. After that request, the client will be subscribed to receive asynchronous events: BalanceUpdated

```
{
  "sid": "4_eb0bf169cbb477b6f8430000015cbe83_fwRFDR4FWFZTV142PiQUCQ4H",
  "date": 1496750162302,
  "class": "EnterLobby",
  "rid": 1,
  "serverId": 1,
  "lang": "en",
  "mode": "real",
  "noFRB": false,
  "gameId": 779,
  "tournamentId": 123444,
  "bonusId": 56767,
  "battlegroundBuyIn": 100
}
```

Request parameters:

sid

session identifier (send as request parameter on start lobby client)

rid

unique request id

serverId

serverId for connect (send as request parameter on start lobby client)

mode

money type: free, real, bonus

gameId

game identifier (send as request parameter on start lobby client)

noFRB

true if server should ignore active FRBs

tournamentId

tournament identifier, optional. (send as request parameter on start lobby client for tournament play)

bonusId

cash bonus identifier, optional, not null only for cash bonus mode (send as request parameter on start lobby client)

battlegroundBuyIn

selected buyIn amount for battleground mode

If session not found, invalid or expired server return

```
{
"code": 3, "msg": "Invalid session", "date": 1496750162288, "class": "Error", "rid": 1
}
```

If selected nickname contains an obscene word, server return:

```
{
  "code": 1000,
  "msg": "Illegal nickname",
  "date": 1496750162288,
  "class": "Error",
  "rid": 1
}
```

If session exist and enter to lobby is success, server return:

```
{
  "players": 600,
  "nickname": "Taras70",
```

```
  "nicknameEditable": true,
  "balance": 20000,
  "currency": { "code": "USD", "symbol": "$" },
  "rankPonts": 200,
  "roomId": 20,
  "avatar": {"borderStyle": 1, "hero": 1, "background": 2},
  "borders": [0, 1, 2, 3],
  "heroes": [0, 1, 2, 3],
  "backgrounds": [0, 1, 2, 3],
  "frBonusInfo": {
    "id": 23142,
    "awardDate": 1533284885209,
    "expirationDate": 1533294885209,
    "totalShots": 500,
    "currentShots": 350,
    "winSum": 20000,
    "stake": 1
  },
  "cashBonusInfo": {
    "id": 23142,
    "awardDate": 1533284885209,
    "expirationDate": 1533294885209,
    "balance": 1000,
    "amount": 10000,
    "amountToRelease": 200,
    "status": "ACTIVE"
  },
  "tournamentInfo": {
    "id": 23142,
    "name": "New Year's tournament",
    "state": "ACTIVE",
    "startDate": 1533284885209,
    "endDate": 1533284885209,
    "balance": 10000,
    "buyInPrice": 1000,
    "buyInAmount": 10000,
    "reBuyAllowed": true,
    "reBuyPrice": 500,
    "reBuyAmount": 10000,
    "reBuyCount": 1,
    "reBuyLimit": -1,
    "resetBalanceAfterRebuy": false
  },
  "battlegroundInfo": {
     "buyIns": [100, 200, 500, 1000, 2000, 5000],
     "alreadySeatRoomId": 200133,
     "startGameUrl":
"https://host/websocket/mpgame?SID=3453&serverId=1&lang=en&roomId=200133",
     "roomState": "WAIT",
     "buyInConfirmed": true
   }
```

```
  "showRefreshBalanceButton": true,
  "rounds": 15,
  "xp": 123,
  "xpPrev": 100,
  "xpNext": 200,
  "place": 10,
  "disableTooltips": false,
  "nicknameGlyphs": "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890",
  "stakesReserve": 300,
  "stakesLimit": 100,
  "alreadySitInStake": 2.0,
  "disableLeaderboard": false,
  "date": 1496908464148,
  "date": 1587381071243,
  "rid": 1,
  "class": "EnterLobbyResponse"
}
```

Response parameters:

players

total players logged in lobby

nickname

player nickname, For new player nickname is null. See 'CheckNicknameAvailability' and 'ChangeNickName' requests.

nicknameEditable

true if nickName may be changed, see ChangeNickname request

balance

current player balance in cents (this is casino side balance, ALL available money). For tournament/cashbonus modes equals to tournament.balance and cashbonus.balance

currency

player currency. Further in this document, all money in player currency

rankPoints

total number of points received by the player, see 4.3 section is spec.

avatar:borderStyle

current avatar borderStyle identifier.

avatar:hero

current avatar hero identifier.

avatar:background

current avatar background identifier.

borders, heroes, backgrounds

- list available avatar parts, see 'ChangeAvatar' request for customize avatar properties.

level

Current player's level

rid

unique request id

roomId

roomId if player already seat in room; -1 if not. If a player occupies a place in a room, the lobby client must prompt the player to continue the game. If the player agrees, the game is started for the given room (send request "GetStartGameUrl" and open game client window)

frBonusInfo

FRB only parameter, always null for real/free/cashbonus/tournament.

cashBonusInfo

CashBonus only parameter, always null for real/free/frbonus/tournament.

cashBonusInfo:balance

current balance.

cashBonusInfo:amount

initial bonus balance

cashBonusInfo:amountToRelease

amount of bets required to release the bonus

cashBonusInfo:status

current bonus status (ACTIVE, RELEASED, LOST, CANCELLED, EXPIRED, CLOSED)

tournamentInfo

tournament only parameter, always null for real/free/frbonus/cashbonus

tournamentInfo:id

unique tournament identifier

tournamentInfo:name

tournament name (may be localized)

tournamentInfo:state

current tournament status (READY, ACTIVE, FINISHED, CANCELED)

tournamentInfo:balance

current player balance in tournament

tournamentInfo:buyInPrice

price of participation in tournament

tournamentInfo:buyInAmount

amount of tournament money which player receives at start of tournament

tournamentInfo:reBuyAllowed

flag indicating that re-buys allowed

tournamentInfo:reBuyPrice

price of purchase of additional tournament money

tournamentInfo:reBuyAmount

amount of tournament money which player receives after re-buy

tournamentInfo:reBuyCount

current re-buys count

tournamentInfo:reBuyLimit

re-buys limit, (-1 for unlimited)

tournamentInfo:resetBalanceAfterRebuy

after re-buy all balance reset to re-buy amount

disableTooltips

enabled/disabled tooltips for player. (see 'ChangeToolTips' request)

stakesReserve

number of stakes to be transferred from balance

stakesLimit

limit of stakes after which you need to make additional buyIn

alreadySitInStake

stake if player seat in room (see roomId), 0 if not.

disableLeaderboard

boolean flag, if true - leaderboard functionality in client should be disabled

nicknameGlyphs

symbols which could be used for nicknames

battlegroundInfo

info about available battlegrounds

battlegroundInfo:buyIns

buyIns available

battlegroundInfo:alreadySeatRoomId

roomId if the player already has incomplete (paid) battleground round

battlegroundInfo:startGameUrl

url must be used for start game client. If this parameter specified, need immediate open client (see specification about explanatory message before open).

battlegroundInfo:roomState

room state if player open room, possible values: WAIT, PLAY, QUALIFY, CLOSED

battlegroundInfo:buyInConfirmed

buyIn confirmed and room open

1.3 CheckNicknameAvailability

After entering the lobby, the player must choose a nickname that will be visible to all other players. To check whether the selected nickname is available, this request is available.

```
{
  "nickname": "Taras",
  "date": 1531731629785,
  "rid": 234,
  "class": "CheckNicknameAvailability"
}
```

Success response:

```
{
  "date": 1531731629785,
  "rid": 1,
  "class": "Ok"
}
```

Error responses:

```
{
  "code": 1015,
  "msg": "Nickname not available",
  "date": 1531731629785,
  "rid":1,
  "class": "Error"
}
{
  "code": 1000,
  "msg": "Illegal nickname",
  "date": 1531731629785,
  "rid": 1,
  "class": "Error"
}
```

1.4 ChangeNickname

```
{
  "nickname": "Taras",
  "date": 1531731629785,
  "rid": 234,
  "class": "ChangeNickname"
}
```

Success response:

```
{
  "date": 1531731629785,
  "rid": 1,
  "class": "Ok"
}
```

Error responses:

```
{
  "code": 1016,
  "msg": "Nickname not available",
  "date": 1531731629785,
  "rid": 1,
  "class": "Error"
}
{
  "code": 1000,
  "msg": "Illegal nickname",
  "date": 1531731629785,
  "rid": 1,
  "class": "Error"
}
{
  "code": 1028,
  "msg": "Not allowed change nickname",
  "date": 1531731629785,
  "rid": 1,
  "class": "Error"
}
```

1.5 ChangeAvatar

```
{
  "borderStyle": 1,
  "hero": 2,
  "background": 3,
  "date": 1531731629785,
  "rid": 234,
  "class": "ChangeAvatar"
}
```

Success response:

```
{
  "date": 1531731629785,
  "rid": 1,
  "class":
  "Ok"
}
[source,json]
.Error responses:
{
  "code": 1017,
  "msg": "Avatar part not available",
  "date": 1531731629785,
  "rid": 1,
  "class": "Error"
}
```

1.6 GetRoomInfo

When the user selects a particular game room, the client requests the server for detailed information about the selected room by a request like:

```
{
  "roomId": 1,
  "rid": 5,
  "date": 1496921495447,
  "class": "GetRoomInfo"
}
```

Request parameters:

roomId

unique room id

rid

unique request id

Response:

```
{
  "roomId": 1,
  "name": "VIP#1",
  "maxSeats": 100,
  "minStake": 0.01,
  "maxStake": 1000.00,
  "currentMultiplier": 2.7,
  "stakes": [
     { "amount": 0.05, "multiplier": 1.5, "autoPlay": true, "win": 0.07 },
```

```
      { "amount": 1.00, "multiplier": 2.5, "autoPlay": true, "win": 2.50 },
      { "amount": 10.00, "autoPlay": false }
    ],
    "state": "PLAY",
    "seats": [
      {
        "id": 1,
        "nickname": "Taras",
        "stakes": [
          { "amount": 0.05, "multiplier": 1.5, "autoPlay": true, "win": 0.07 },
          { "amount": 1.00, "multiplier": 2.5, "autoPlay": true, "win": 2.50 },
          { "amount": 10.00, "multiplier": 3.0, "autoPlay": true }
        ]
      },
      {
        "id": 2,
        "nickname": "Bulba",
        "stakes": [
          { "amount": 0.10, "multiplier": 1.5, "autoPlay": true, "win": 0.15 },
        ]
      },
      {
        "id": 3,
        "nickname": "Mikola",
        "stakes": [
          { "amount": 1.00, "multiplier": 2.5, "autoPlay": true, "win": 2.50 },
          { "amount": 10.00, "multiplier": 3.1, "autoPlay": true }
        ]
      }
    ],
    "ttnx": 7,
    "alreadySitInNumber": -1,
    "alreadySitInBalance": 0,
    "alreadySitInWin": 0.25,
    "subround": "BASE",
    "roundId": 450122,
    "cashBonusInfo": {
      "id": 23142,
      "awardDate": 1533284885209,
      "expirationDate": 1533294885209,
      "balance": 1000,
      "amount": 10000,
      "amountToRelease": 200,
      "status": "ACTIVE"
    },
    "tournamentInfo": {
      "id": 23142,
      "name": "New Year's tournament",
      "state": "ACTIVE",
      "startDate": 1533284885209,
      "endDate": 1533284885209,
```

```
      "balance": 10000,
      "buyInPrice": 500,
      "buyInAmount": 10000,
      "reBuyAllowed": true,
      "reBuyPrice": 500,
      "reBuyAmount": 100000,
      "reBuyCount": 1,
      "reBuyLimit": -1,
      "resetBalanceAfterRebuy": false
    },
    "battlegroundInfo" {
      "buyIn" : 1000,
      "buyInConfirmed": true,
      "timeToStart" : 30,
      "kingsOfHill": [1,2,3],
      "score": 4620,
      "rank" : 3,
      "pot": 3960,
      "potTaxPercent": 5
    },
    "date": 1497004331658,
    "rid": 5,
    "class": "GetRoomInfoResponse"
  }
```

Where:

rid

request id from “GetRoomInfo”

roomId

unique room id

name

room name

maxSeats

max seats in room

minStake

minimal allowed stake in cents (type – float)

maxStake

maximal allowed stake in cents (type – float)

currentMultiplier

current multiplier, zero in round not started.

playerStake

player stake (shot cost) in cents (type – float).

stakes

current player stakes, may be multiple, max 3 astronaut

stakes::amount

stake amount in cents (type – float)

stakes::multiplier

stake multiplier, may be zero if not autoplay and win not defined

stakes::autoPlay

autoPlay mode: boolean, mandatory

stakes::win

win amount in cents (type – float), may be not defined if rocket not crash and astronaut did not catapults. Zero for loss.

state

current room state. Possible values: WAIT, PLAY, QUALIFY, CLOSED

seats

other players info

seats::id

seat identifier, deprecated. Always 0. Use nickname for identify seat

seats::nickname

seat nickname

seats::stakes

player stakes info

ttnx

rough time to next state in seconds, -1 for unknown (ttnx -Time To NeXt).

alreadySitInNumber

seat number; -1 if player not seat in room;

alreadySitInBalance

current player balance if player seat in room; 0 if player not seat

alreadySitInWin

player's win in current round

cashBonusInfo

same as EnterLobbyResponse.cashBonusInfo

tournamentInfo

same as EnterLobbyResponse.tournamentInfo

battlegroundInfo

battleground info, always null for other modes

battlegroundInfo:buyIn

buyIn amount for join

battlegroundInfo:buyInConfirmed

true if buyIn confirmed by player

battlegroundInfo:timeToStart

time to start next battleground round in seconds

battlegroundInfo:kingsOfHill

Kings of the Hill seat number ids

battlegroundInfo:score

current player score (ath this moment always total win amount)

battlegroundInfo:rank

current player rank

battlegroundInfo:pot

pot amount

battlegroundInfo:potTaxPercent

casino tax percentage of pot amount

1.7 BalanceUpdated

This message sent by server side after changing player balance. Balance is checked every 15 seconds on server side.

Response

```
{
  "class": "BalanceUpdated",
  "date": 1496748898812,
  "rid": -1,
  "balance": 1236600,
  "serverAmmo": 293
}
```

1.8 GetTime

Sended from client each N seconds. N(MQ_TIMER_FREQ) in getParams() in template.jsp.

Request

```
{
  "class": "GetTime",
  "date": 1496748898812,
  "rid": 10
}
.Response
{
  "class": "TimeUpdated",
  "date": 1496748898813,
  "rid": 10
}
```

1.9 TournamentStateChanged

Server sends this message after tournament status changed

```
{
  "id": 323453,
  "oldState": "ACTIVE",
  "newState": "EXPIRED",
  "reason": "Tournament time is up",
  "date": 1531731629785,
  "rid": -1,
  "class": "TournamentStateChanged"
}
```

Where:

id

tournament id

reason

additional info about change state, may be empty.

oldState

READY, ACTIVE

newState

ACTIVE, FINISHED, CANCELED

1.10 BonusStatusChanged

Server sends this message after bonus status changed

```
{
  "id": 345345,
  "type": "FRB",
  "oldStatus": "ACTIVE",
  "newStatus": "FINISHED",
  "reason": "Bonus expired",
  "date": 1531731629785,
  "rid": -1,
  "class": "BonusStatusChanged"
}
```

Where:

id

bonus id

type

bonus type, may be FRB or CASHBONUS

reason

additional info about change state, may be empty.

oldStatus

ACTIVE

newStatus

RELEASED, LOST, CANCELLED, EXPIRED, CLOSED

1.11 OpenRoom

After this request is completed, the client goes into observer mode and begins to receive all necessary information about the game.

```
{
  "roomId": -1,
  "lang": "en",
  "sid": "eb0bf169cbb477b6f8430000015cbe83",
  "date": 1497157072066,
  "rid": 5,
  "class": "OpenRoom"
}
```

Request parameters:

roomId

always -1 for crash games

lang

preferred language

rid

unique request id

sid

session identifier send as request parameter on start game client.

Response:

GetRoomInfoResponse. See paragraph 1.6

If game state is “PLAY” client immediately receives message:


```
{
  "subround": "BASE",
  "startTime": 0,
  "state": "PLAY",
  "seats": [
    {
      "id": 1,
      "nickname": "Taras",
      "stakes": [
         { "amount": 0.05, "multiplier": 1.5, "autoPlay": true, "win": 0.07 },
         { "amount": 1.00, "multiplier": 2.5, "autoPlay": true, "win": 2.50 },
         { "amount": 10.00, "multiplier": 3.0, "autoPlay": true }
      ]
    },
    {
      "id": 2,
      "nickname": "Bulba",
      "stakes": [
         { "amount": 0.10, "multiplier": 1.5, "autoPlay": true, "win": 0.15 },
      ]
    },
    {
      "id": 3,
      "nickname": "Mikola",
      "stakes": [
         { "amount": 1.00, "multiplier": 2.5, "autoPlay": true, "win": 2.50 },
         { "amount": 10.00, "multiplier": 3.1, "autoPlay": true }
      ]
    }
  ],
  "roundId": 450063,
  "cashBonusInfo": {
    "id": 23142,
    "awardDate": 1533284885209,
    "expirationDate": 1533294885209,
    "balance": 1000,
```

```
    "amount": 10000,
    "amountToRelease": 200,
    "status": "ACTIVE"
  },
  "tournamentInfo": {
    "id": 23142,
    "name": "New Year's tournament",
    "state": "ACTIVE",
    "startDate": 1533284885209,
    "endDate": 1533284885209,
    "balance": 10000,
    "buyInPrice": 500,
    "buyInAmount": 10000,
    "reBuyAllowed": true,
    "reBuyPrice": 500,
    "reBuyAmount": 100000,
    "reBuyCount": 1,
    "reBuyLimit": -1,
    "resetBalanceAfterRebuy": false
  },
  "battlegroundInfo" {
      "buyIn" : 1000,
      "buyInConfirmed": true,
      "timeToStart" : 0,
      "kingsOfHill": [-1],
      "score": 0,
      "rank" : 0,
      "pot": 3960
  }
  "date": 1587912885176,
  "rid": 5,
  "class": "FullGameInfo"
}
```

1.11 CloseRoom

This request must be send to server before close client. This required for release all server resources allocated to client.

```
{
  "roomId": 1,
  "date": 1497159892576,
  "rid": 5,
  "class": "CloseRoom"
}
```

Request parameters:

roomId

unique room id

rid

unique request id

Response:

```
{
  "date": 1497154639465,
  "rid": 1,
  "class": "Ok"
}
```

1.12 SitIn

The request is sent by the client to seat in the room.After the successful execution of this request, the user moves from the observer state to the player's state and additional actions available to him. In battleground mode sitIn must be sended only after accept buyIn.If buyIn not accepted need send CloseRoom message.

```
{
  "date": 1497160498696,
  "rid": 1,
  "stake": 5,
  "class": "SitIn"
}
```

Request parameters:

stake

stake in cents. This parameter must be 0 for re-sitIn

Response:

```
{
  "id": 4,
  "nickname": "Andrey",
  "enterDate": 1497160789104,
  "balance": 15000,
  "avatar": {
    "borderStyle": 1,
    "hero": 1,
    "background": 10
  },
  "showRefreshBalanceButton": true,
  "level": 8,
  "frbMode": false,
  "frbBalance": 0,
  "mode": "FRB",
  "maxMultiplier": 1000000.00,
  "date": 1497160789056,
  "rid": 2,
  "class": "SitInResponse"
}
```

Where:

rid

request id from “SitIn”

id

seat number

nickname

seat nickname

balance

current player balance

avatar

seat avatar

showRefreshBalanceButton

need show "Refresh Balance" button

frbMode

deprecated parameter, see 'mode' (will be deleted after client fix)

frbBalance

deprecated parameter (will be deleted after client fix)

mode

game mode, possible values FRB, CASHBONUS, REAL,TOURNAMENT

maxMultiplier

max allowed multiplier

Other clients also receive this response, but with rid=-1.

1.13 SitOut

The request is sent by the client in order to seat out from the room. After that the user goes into observer mode. For tournaments, cashbonus, frbonus modes, not required send a SitOut request, this will be done automatically and the client will receive a SitOutResponse message.

```
{
  "date": 1497161230259,
  "rid": 1,
  "class": "SitOut"
}
```

Response:

```
{
  "id": 4,
  "nickname": "Andrey",
  "outDate": 1497161341687,
  "nextRoomId": 32445,
  "hasNextFrb": true,
  "date": 1497161341653,
  "rid": 1,
  "class": "SitOutResponse"
}
```

Where:

rid

request id from “SitOut”

id

seat identifier, deprecated. Always 0. Use nickname for identify seat

nickname

seat nickname

nextRoomId

roomId for sitIn to real room (after cashBonus or tournament), may be -1 if there is no suitable room

hasNextFrb

after cashbonus or frb completed, this flag indicating that exist new FRBonus

Other clients also receive this response, but with rid=-1.

1.14 GameStateChanged

This message sent from server to all clients on each game state change.

```
{
  "state": "PLAY",
  "ttnx": 600,
  "roundId": 12345,
  "date": 1497171419692,
  "rid": 1,
  "class": "GameStateChanged"
}
```

Where:

state

current room state. Possible values: WAIT, PLAY, QUALIFY, CLOSED, BUY_IN

ttnx

rough time to next state in seconds, -1 for unknown

roundId

current round id

rid

request id, always -1

1.15 GetFullGameInfo

```
This request can be sent by the client if connection lost and need refresh all room i
nfo.
{
  "date": 1497178236042,
  "rid": 1,
  "class": "GetFullGameInfo"
}
```

Response: same as FullGameInfo in paragraph 2.2

1.16 RoundResult

After round finished all clients receive message “GameStateChanged” described in paragraph 2.6 with state=QUALIFY. After qualification completed all clients receive round result in message:

```
{
  "winAmount": 500.5,
  "balance": 12000,
  "roundId": 1090142,
  "seats": [
    {
      "id": 1,
      "nickname": "Taras",
      "stakes": [
         { "amount": 0.05, "multiplier": 1.5, "autoPlay": true, "win": 0.07 },
         { "amount": 1.00, "multiplier": 2.5, "autoPlay": true, "win": 2.50 },
         { "amount": 10.00, "multiplier": 3.0, "autoPlay": true, "win": 0.00 }
      ]
    },
    {
      "id": 2,
      "nickname": "Bulba",
      "stakes": [
         { "amount": 0.10, "multiplier": 1.5, "autoPlay": true, "win": 0.15 },
      ]
    },
    {
      "id": 3,
      "nickname": "Mikola",
      "stakes": [
         { "amount": 1.00, "multiplier": 2.5, "autoPlay": true, "win": 2.50 },
         { "amount": 10.00, "multiplier": 3.1, "autoPlay": true,"win": 0.00 }
      ]
    }
  ],
  "realWinAmount": 10.0
  "date": 1497179064748,
  "rid": -1,
  "class": "RoundResult",
  "battlegroundRoundResult": [
    {
       "id": 0,
       "score": 12324,
       "rank": 1,
       "pot": 3960
    },
    {
       "id": 1,
       "score": 11324,
       "rank": 2,
```

```
        "pot": 0
    },
    {
        "id": 2,
        "score": 10324,
        "rank": 3,
        "pot": 0
    },
    {
        "id": 5,
        "score": 1324,
        "rank": 4,
        "pot": 0
    }
  ]
}
```

Where:

winAmount

amount won in this round (type - float)

balance

current player balance

battlegroundRoundResult

battleground mode leaderboard info

1.17 FRBEnded

After FRB completed, server send FRBEnded message. Client must display close reason (Completed, Canceled, Expired) and win sum. After close message box, need close client and resend from lobby messages: EnterLobby, GetRooms

Response

```
{
  "winSum":1200,
  "closeReason":"Completed",
  "hasNextFrb":true
  "date":1533285929327,
  "rid":-1,
  "class":"FRBEnded"
}
```

1.18 ReBuy

Request

```
{
  "date": 1496748898812,
  "class": "ReBuy",
  "rid": 5
}
```

Response

```
{
  "date": 1496748898812,
  "class": "ReBuyResponse",
  "rid": 5,
  "ammoAmount": 10000,
  "balance": 12345,
  "rebuyCount": 2
}
```

1.19 CancelBattlegroundRound

Message received to game client if battleground round canceled.Cliet must display meassage and return to lobby.

```
{
      "refundedAmount": 1000,
      "reason": NOT_ENOUGH_PLAYERS,
      "class":"CancelBattlegroundRound",
      "rid":9,
      "date":1617613404721
}
```

refundedAmount

returned buyIn, may be 0 if played not make buyIn

reason

cancel reason. enum with possible values: NOT_ENOUGH_PLAYERS, SERVER_FAILURE

1.20 KingOfHillChanged

Message sended after change King of the Hill

```
{
      "newKings": [1,2,3],
      "class":"KingOfHillChanged",
      "rid":9,
      "date":1617613404721
}
```

newKings

new King seatIds

1.21 ConfirmBattlegroundBuyIn

Message must be sended game client if player confirm buyIn for next battleground

```
{
      "class":"ConfirmBattlegroundBuyIn",
      "rid":9,
      "date":1617613404721
}
```

Success response:

```
{
  "date": 1531731629785,
  "rid": 9,
  "class": "Ok"
}
```

1.22 Crash Bet

The request is sent by the client if need make bet in crash game.

```
{
  "crashBetAmount": 100,
  "multiplier": 1.35,
  "autoPlay": false,
  "date": 1497174203319,
  "rid": 10,
  "class": "CrashBet"
}
```

crashBetAmount

amount of crash bet (cents)

multiplier

crash multiplier of bet

autoPlay

need auto collect on this multiplier.

Response:

```
{
  "crashBetAmount": 120,
  "balance": 15000,
  "date": 1497174620105,
  "crashBetKey": 1497174620105_0,
  "rid": 10,
  "class": "CrashBetResponse"
}
```

Where:

rid

request id from “BuyIn” request

crashBetAmount

amount of crash bet (cents)

crashBetKey

key of crash bet (will be needed later in cancel of bet)

balance

current balance in cents

1.23 Cancel Crash Bet

The request is sent by the client if need cancel bet in crash game. Player will get win with current multiplier.

```
{
  "crashBetId": 1497174620105_0,
  "placeNewBet": false,
  "date": 1497174203319,
  "rid": 10,
  "class": "CrashCancelBet"
}
```

crashBetId

id of crash bet (string)

placeNewBet

true for cancel bet before new placeBet, false if round started and need jump from rocket

Response:

```
{
  "currentMult": 2.3,
  "nickname": "User8677",
  "seatWin": 1550,
  "date": 1497174620105,
   "rid": 10,
  "class": "CrashCancelBetResponse"
}
```

1.24 CrashStateInfo

The response is sent by the server to all players if multipliers was changed in room.

```
{
  "currentMult": 3.1,
  "timeSpeedMult" : 4.0,
  "date": 1497174203319,
  "rid": 10,
  "class": "CrashStateInfo"
}
```

Where:

rid

request id from “BuyIn” request

crashBetAmount

amount of crash bet (cents)

crashBetKey

key of crash bet (will be needed later in cancel of bet)

balance

current balance in cents

timeSpeedMult

standard 1.0 in SP version, changed in btg mode when all bets ejected for speed up rocket

1.25 CrashCancelAutoEject

The request is sent by the client if need cancel autoeject bet in crash game.

```
{
  "betId": 1497174620105_0,
  "date": 1497174203319,
  "rid": 10,
  "class": "CrashCancelAutoEject"
}
```

crashBetId

id of crash bet (string)

Response:

```
{
  "nickname": "User3453",
  "betId": 1497174620105_0,
  "date": 1497174620105,
  "rid": 10,
  "class": "CrashCancelAutoEjectResponse"
}
```

1.26 CrashChangeAutoEject

The request is sent by the client if need change autoeject multiplier in crash game.

```
{
  "betId": 1497174620105_0,
  "date": 1497174203319,
  "multiplier": 1.35,
  "rid": 10,
  "class": "CrashChangeAutoEject"
}
```

crashBetId

id of crash bet (string)

Response:

```
{
  "nickname": "User3453",
  "betId": 1497174620105_0,
  "date": 1497174620105,
  "rid": 10,
  "class": "CrashChangeAutoEjectResponse"
}
```

1.27 Reserve bet extension

This extension is used for licensees who have not implemented extended CW protocol (realBet, realWin).

Server sends a boolean parameter CW_SEND_REAL_BET_WIN in getParams() in template.jsp to Client. If the parameter equals to false client should use "Reserve bet extension".

In this extension there is an additional state BuyIn State.

- Client starts game as usual, when server receives Bet:
  - Server checks if Bet is correct
  - Saves buyIn with reserved flag
  - Does not process buyIn
  - Sends CrashBetResponse **without** changing balance

Response:

```
{
  "crashBetAmount": 120,
  "balance": 15000,
  "date": 1497174620105,
  "crashBetKey": 1497174620105_0,
  "rid": 10,
  "class": "CrashBetResponse"
}
```

- When server receives a cancel Bet request
  - Server checks the request, remove the bet and sends CrashCancelBetResponse

Response:

```
{
  "currentMult": 2.3,
  "nickname": "User8677",
  "seatWin": 1550,
  "date": 1497174620105,
   "rid": 10,
  "class": "CrashCancelBetResponse"
}
```

- When game reaches the last 5 seconds of Waiting state, state is changed to BuyIn state. Server sends GameStateChanged message

```
{
  "state": "BUYIN",
  "ttnx": 5,
  "roundId": 12345,
  "date": 1497171419692,
  "rid": 1,
  "class": "GameStateChanged"
}
```

- Server processes buyIn for sum of all reserved bets
- In case succeed buyIns for all players after 5 seconds state changed to Play state. Server sends GameStateChanged message.

```
{
  "state": "PLAY",
  "ttnx": 600,
  "roundId": 12345,
  "date": 1497171419692,
  "rid": 1,
  "class": "GameStateChanged"
}
```

- For failed bets server sends CrashAllBetsRejectedResponse. All bets for player should be canceled.

Response:

```
{
  "date": 1497174620105,
  "rid": -1,
  "seatId" : 0,
  "name": "User8677",
  "errorCode": 100300,
  "errorMessage": "BuyIn failed, reason: insufficient funds",
  "class": "CrashAllBetsRejectedResponse"
}
```

In case there are pending bets we wait all responses from GS and after that change state to Play.

3. Error codes

Error code	Error name	Description
1	INTERNAL_ERROR	Unexpected error, please report it to BSG Server team
2	SERVER_SHUTDOWN	
3	INVALID_SESSION	Please use another token to login into the game
4	DEPRECATED_REQUEST	
5	FOUND_PENDING_OPERATION	This user is unable to login due to problems with his account
6	CLOSE_FRB_IN_PROGRESS	Message is removed from the game
7	UNKNOWN_GAME_ID	Wrong game id
8	FOUND_OPEN_LOBBY_SESSION	Found other open lobby for EnterLobby (and game mode conflict).
1000	ILLEGAL_NICKNAME	Nickname contains banned patterns, please use another one
1001	BAD_REQUEST	Message received from the client has invalid format
1002	NOT_LOGGED_IN	Please restart game
1003	ROOM_NOT_FOUND	Please select another room
1004	TOO_MANY_OBSERVERS	Room is full, please select another one
1005	ROOM_MOVED	Room moved to another server
1006	TOO_MANY_PLAYER	All 6 slots for players are taken, you can observe game until one is freed or pick another one
1007	NOT_SEATER	You should perform SitIn operation before you can make actions in this room
1008	NOT_ENOUGH_MONEY	There is not enough money for purchase
1009	BAD_STAKE	Invalid stake size is provided by client
1010	BAD_BUYIN	Invalid BuyIn request is send by client
1011	CHANGE_STAKE_NOT_ALLOWED	Change of stake is forbidden at this moment (not used)

Error code	Error name	Description
1012	ROOM_NOT_OPEN	You should send OpenRoom request first
1013	ROUND_NOT_STARTED	You need to wait until round is started until you can make actions in this room
1014	NEED_SITOUT	You should perform SitOut request
1015	WRONG_WEAPON	You are trying to perform shot using inactive weapon
1016	NICKNAME_NOT_AVAILABLE	Nickname is already taken by another player, please pick another one
1017	AVATAR_PART_NOT_AVAILABLE	You are trying to use avatar which is not available for this account yet
1018	NOT_ENOUGH_BULLETS	You need to purchase bullets before you can shoot
1019	REQUEST_FREQ_LIMIT_EXCEEDED	You are trying to send requests too fast, please limit rate
1020	QUEST_IS_NOT_COMPLETED	You can’t collect quest which is not completed
1021	BUYIN_NOT_ALLOWED	You can perform BuyIn in this game state
1022	QUEST_ALREADY_COLLECTED	You already collected this quest
1023	WRONG_MINE_COORDINATES	You are trying to place mine on invalid location
1024	BOSS_ROUND_STARTED	Not used
1025	WRONG_UNPLAYED_FREE_SHOTS	Not used
1026	NOT_FATAL_BAD_BUYIN	Buy in is failed, but you can proceed with playing
1027	NOT_ALLOWED_CHANGE_BET	This message is sent by server when changing of bet levels is prohibited to avoid cheats with RTP
1028	NOT_ALLOWED_CHANGE_NICKNAME	Change nickname not allowed
1030	NOT_ALLOWED_SITIN	Sit in is not allowed during Qualify game state, client should wait and resend request

5. Special Weapons

Title	Type Id
Shotgun	0
Grenade	1
Machine Gun	2
Laser	3
Plasma Rifle	4
Mine Launcher	5
Railgun	6
Artillery Strike	7
Rocket Launcher	8
Flamethrower	9
Cryogun	10
Rapid Fire Pistol	11